

Lesson 1.1: Why We Need Oxygen – The Body's Fuel

What You'll Learn

By the end of this lesson, you'll understand why oxygen is so important for your child's body.

The Content

Think of oxygen as fuel for your body.

Just like a car needs gas to run, every cell in your body needs oxygen to make energy. But oxygen isn't the energy itself – it's used to make the energy.

Inside every cell, there are tiny power plants that use oxygen to create a special molecule called ATP (say: A-T-P). ATP stands for adenosine triphosphate (say: ah-DEN-oh-seen try-FOS-fate), but you can just remember it as ATP. ATP is the actual fuel that powers everything your cells do.

When cells use oxygen to make ATP, they also create carbon dioxide as a waste product. This carbon dioxide needs to leave the body, which is why we breathe it out.

The parts of your body that do the most work need the most energy. Your brain is always thinking and sending signals. Your heart is always beating. Your muscles are always moving. Your kidneys, liver, and other organs are always working. All of these hard-working parts need a lot of ATP, which means they need a steady supply of oxygen.

Where does oxygen come from?

Oxygen is in the air all around us. But air isn't just oxygen – it's a mixture of different gases:

- **Nitrogen:** about 78% of air
- **Oxygen:** about 21% of air
- **Carbon dioxide:** less than 1% of air
- **Other gases:** less than 1% of air

When we breathe, our bodies go through a process called gas exchange. We take in oxygen from the air and release carbon dioxide back into the air. This happens automatically with every breath.

What happens without enough oxygen?

Without enough oxygen, your body can't make enough ATP energy. This is especially hard on important organs like the brain and heart. They need a steady supply of oxygen to work properly.

When someone has lung disease, the oxygen level in their body can be low. When this happens, the body has to work much harder just to make enough energy to function.

This extra work can make a person:

- Feel more tired than usual
- Have less energy for activities
- Grow more slowly (in children)
- Have trouble doing normal daily tasks

What is SpO2?

You may hear doctors talk about "SpO2" or "oxygen saturation." This is a number that shows how much oxygen is in your child's blood. But to understand this number, we need to talk about how oxygen travels in the blood.

How oxygen travels in the blood

Oxygen doesn't just float freely in the blood. It needs a ride to get to all the cells in the body. This ride comes from special cells called red blood cells.

Inside each red blood cell is a protein called hemoglobin (say: HEE-moh-glow-bin). Think of hemoglobin as a taxi that carries oxygen. Each hemoglobin molecule can pick up four oxygen passengers and deliver them to cells throughout the body.

When we measure SpO2, we're measuring how many of these hemoglobin "taxis" are carrying oxygen. If the SpO2 is 98%, it means that 98% of the hemoglobin in the blood is carrying oxygen.

What do the SpO2 numbers mean?

Doctors use a small clip called a pulse oximeter on your child's finger or toe to check SpO2. It doesn't hurt at all. Here's what the numbers mean:

- **95% or higher** – This is normal. Your body works best when SpO2 is in this range. Most healthy people have an SpO2 between 95% and 100%.

- **90-94%** – This is lower than normal. At these levels, the body is getting oxygen, but it may need some help. Doctors might give supplemental oxygen at this point.

- **85-92%** – At this level, the body can still supply oxygen to the organs, but the heart and lungs have to work much harder to do it. Think of it like running instead of walking to get somewhere – you can do it, but it takes a lot more energy. Over time, when the heart and lungs have to work this hard all the time, it can lead to:

- Poor growth in children
- Tiredness and low energy
- Problems with the heart (because it's working too hard)

- **Below 88%** – This is very serious. At these levels, the body is not getting enough oxygen, and organs like the brain and heart may not work properly. This can be life-threatening and needs immediate medical attention.

This number is one of the most important ways to know if your child is getting enough oxygen.

Key Takeaway: Oxygen is like fuel for the body. Every cell needs it to make energy and stay healthy.
